

Module - 2

Bivariate Data and Multivariate Data:-

* Bivariate data involves 2 variables and focuses on finding relationships b/w them.

Ex: Temperature in a shop and sales data.

Temp ($^{\circ}\text{C}$)	Sales of sweaters
5	200
10	150
20	140

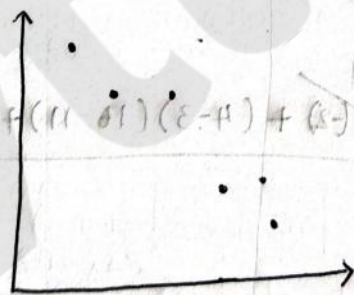
To analyze relationships, a scatter plot is used.

* It helps visualize trends, differences and relationships b/w 2 variables.

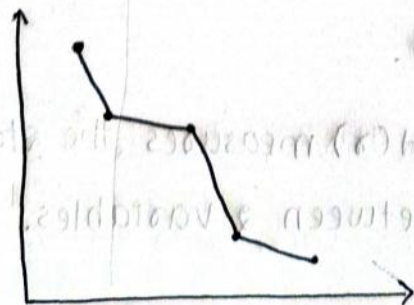
* Shows strength, shape, direction and presence of outliers.

* Useful for exploratory data before calculating correlation (or) regression.

Ex:-



Line graphs are similar to scatter plots.



Bivariate Statistics:-

*> Covariance and Correlation are examples of bivariate statistics.

Covariance:-

*> Denoted by $\text{cov}(X, Y)$

*> It is used to measure the variance b/w two dimensions.

Formula:-

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(X))(y_i - E(Y))$$

where, x_i and $y_i \rightarrow$ data values from X and Y .

$E(X)$ and $E(Y) \rightarrow$ mean values of x_i and y_i

$N \rightarrow$ number of given data.

Q> Find the covariance of data, $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 4, 9, 16, 25\}$

Sol:- Mean $(X) = E(X) = \frac{15}{3} = 5$

and $E(Y) = \frac{55}{5} = 11$

$$\therefore \text{cov}(X, Y) = \frac{(1-3)(1-11) + (2-3)(4-11) + \cancel{(3-3)(9-11)} + (4-3)(16-11) + (5-3)(25-11)}{5}$$

$$\boxed{\text{cov}(X, Y) = 12}$$

Correlation:- (denoted by r).

*> Pearson correlation coefficient (r) measures the strength and direction of the linear relationship between 2 variables.

Formula:-

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

$\sigma_x, \sigma_y \rightarrow$ ^{population} standard deviations.
 $\Rightarrow \sigma = \sqrt{\frac{(x_i - \bar{x})^2}{n}}$

Note:- If $\rho > 0$, X and Y increase together

If $\rho < 0$, one increases while the other decreases.

If $\rho = 0$, they are independent.

Q) Find the correlation for data, $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 4, 9, 16, 25\}$

Sol:- $E(X) = \mu_x = 3$ and $E(Y) = \mu_y = 11$

$$\sigma_x = \sqrt{\frac{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2}{5}}$$

$$\sigma_x = 1.4142$$

$$\sigma_y = \sqrt{\frac{(1-11)^2 + (4-11)^2 + (9-11)^2 + (16-11)^2 + (25-11)^2}{5}}$$

$$\sigma_y = 8.6486$$

$$\text{cov}(X, Y) = 12$$

$$\therefore \rho = \frac{12}{1.4142 \times 8.6486}$$

$$\rho = 0.984$$

Multivariate Statistics:-

* Multivariate data involves more than two variables and is commonly used in ML.

* Similar to bivariate data but with multiple dependent variables.

* Common analysis techniques:- Regression analysis, Principal Component Analysis (PCA), Path analysis

Ex:-

Id	Attribute 1	Attribute 2	Attribute 3
1	1	4	1
2	2	5	2
3	3	6	1

* Example data has 3 attributes, and their mean is called a centroid.

* Variance of multivariate data is represented as a covariance matrix.

Heatmap:-

* A graphical representation of a 2D matrix using colors.

* Darker colors = larger values, lighter colors = smaller values.

* Helps in perceiving patterns easily.

* Ex:- Traffic analysis (high vs low traffic regions)
health data (weight vs health status)

Pairplot:-

* A visualization tool for multivariate data.

* Consists of scatter plots for each pair of variables.

*> Helps in finding relationships and correlations.

Ex:- A random matrix of 3 columns is plotted as a pair plot to show relationships.

Essential mathematics for multivariate data:-

*> ML requires concepts from linear algebra, statistics, probability and information theory.

*> Linear algebra is essential for ML, dealing with linear equations, vectors, matrices and transformations.

Linear Systems and Gaussian Elimination for multivariate data:-

*> A linear system of equations is a group of equations with unknown variables.

Written as $Ax=y$ and solved as $x=A^{-1}y$, if A is non-zero.

*> If a unique solution exists, the system is consistent independent.

If multiple solutions exist, it is consistent dependent. If no solution exists, it is inconsistent.

*> Gaussian Elimination solves large systems using row operations:

i) Convert the system into an augmented matrix

ii) Use a pivot element to eliminate variables in other rows.

iii) Convert to row echelon form.

iv) Use back-substitution to find unknowns.

Q) Given, set of eqns, $2x_1 + 4x_2 = 6$

$4x_1 + 3x_2 = 7$

Solve by Gaussian Elimination method.

Solr Let $A =$ augmented matrix.

$$A = \left[\begin{array}{cc|c} 2 & 4 & 6 \\ 4 & 3 & 7 \end{array} \right]$$

$$A = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 3 & 7 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$R_2 = R_2 - 4R_1$$

$$\Rightarrow A = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -5 & -5 \end{array} \right]$$

$$R_2 \rightarrow R_2 / -5$$

$$\Rightarrow A = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 2R_2$$

$$A = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array} \right]$$

$\therefore$ Now, $x_1 = 1$

$$x_2 = 1$$

Matrix decompositions:-

Matrix decomposition breaks a matrix into smaller parts for easier calculations.

* Eigen decomposition:-

→ A matrix A is decomposed into eigenvalues and eigenvectors:

$$A = Q \Lambda Q^T$$

where $Q \rightarrow$ eigenvector matrix

$\Lambda \rightarrow$ diagonal matrix

$Q^T \rightarrow$ transpose of Q .

LU Decomposition:-

* LU decomposition splits a matrix into:

$$A = LU$$

where $L \rightarrow$ lower triangular matrix

$U \rightarrow$ upper triangular matrix.

Steps:-

i) Start with an identity matrix

ii) Apply Gaussian elimination

iii) Reduce the matrix to get L and U .

iv) Find LU decomposition of given matrix:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix}$$

Sol:-

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix}$$

$$R_2 = R_2 - 3R_1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 3 & 4 & 2 \end{bmatrix} \quad \text{Temp} = A$$

$$R_3 = R_3 - 3R_1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & -2 & -10 \end{bmatrix}$$

$$R_3 = R_3 - \frac{2}{3} R_2$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & 0 & -\frac{10}{3} \end{bmatrix}$$

$$\Rightarrow \boxed{A = L * U}$$

Machine Learning and Importance of Probability and Statistics:-

* ML is closely linked to probability and statistics. Probability helps in understanding data distributions, while statistics assist in data analysis. Many ML datasets follow different probability distributions, making their knowledge essential.

Probability Distributions:-

A probability distribution describes how values of a variable (X) are distributed. It is a mathematical function that determines the probability of different values in a dataset.

Types of probability distributions:-

- 1) Discrete Probability Distribution:- Deals with countable values.
- 2) Continuous Probability Distribution:- Deals with infinite possible values.

Key concepts:-

- * Probability Density Function (PDF):- Determines probability for continuous variables.
- * Cumulative Distribution Function (CDF):- Probability of a value being less than (or) equal to a given number.
- * Probability mass function (PMF):- Used for discrete distributions (equivalent of PDF)

Continuous Probability Distributions:-

1) Normal Distribution (Gaussian distribution):-

- * Bell-shaped curve, most common distribution
- * Mean, median and mode are equal.
- * Used for heights, blood pressure and test scores.

Formula:-

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

* z-score $\Rightarrow$ $z = \frac{x - \mu}{\sigma}$

2) Rectangular (uniform) distribution:-

*> All values in the range $[a, b]$ have equal probability.

*> Used for modelling fair dice rolls, random number generation.

Formula:-

$$P(X=x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

3) Exponential distribution:-

*> Used for modeling time until an event happens (eg. waiting time for a bus).

*> Special case of the Gamma distribution.

Formula:-

$$f(x, \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

*> Mean = $\frac{1}{\lambda}$ and std. dev = $\frac{1}{\lambda}$

Discrete Probability Distribution:-

1) Binomial Distribution:-

*> models the probability of x successes in n trials (eg. coin flips)

*> Each trial has two outcomes: success (p) or failure ($1-p$) $\rightarrow q$.

Formula:-

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

where, $q = 1-p$.

*> Mean = np

variance = npq

2) Poisson Distribution:-

* Models the number of events in a fixed time (eg., number of emails per hour)

* Used when events are independent.

Formula:-

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

* Mean = λ , Standard deviation = $\sqrt{\lambda}$

3) Bernoulli Distribution:-

* Models an experiment with only 2 outcomes: success (1) or failure (0)

* Used in binary classification problems.

* Formula:-

$$f(k;p) = \begin{cases} 1-p, & k=0 \\ p, & k=1 \end{cases}$$

* Mean = p , variance = pq

Density Estimation:-

* Let $X = \{x_1, x_2, \dots, x_n\}$ be a set of observed values from an unknown distribution. Density estimation is the process of estimating the probability density function $p(x)$ from these observations.

* If $p(x)$ is small, x may be an anomaly.

* There are 2 main types:-

1) Parametric Density Estimation

2) Non-Parametric Density Estimation.

Parametric Density Estimation:-

*> Assumes data follows a known probability distribution $p(x|\theta)$, where θ is the parameter set.

Maximum Likelihood Estimation (MLE):-

*> MLE is a fundamental statistical method used to estimate parameters of a probability distribution by maximizing the likelihood function.

*> The idea behind MLE is to choose the parameter θ that makes the observed data $X = \{x_1, x_2, \dots, x_n\}$ most probable under the assumed model.

MLE Definition:-

For a given dataset X , the likelihood $L(X;\theta)$ is defined as:

$$L(X;\theta) = \prod_{i=1}^n p(x_i;\theta)$$

This function represents the probability of observing the given data under a specific choice of parameters θ . The goal is to find θ that maximizes $L(X;\theta)$:

$$\text{max}_{\theta} L(X;\theta)$$

Log-likelihood Function:-

*> Since computing the product of probabilities can lead to very small numbers (causing numerical instability), we take the natural logarithm of the likelihood function. This transforms the problem into

log-likelihood maximization:

$$\max \sum_{i=1}^n \log p(x_i; \theta)$$

Alternatively, we can frame the problem as minimizing the negative log-likelihood (NLL), which is commonly done in optimization tasks

$$\min \sum_{i=1}^n \log p(x_i; \theta)$$

Minimizing the NLL is often preferred in ML, as loss functions in neural networks and statistical models are designed in this form.

MLE in Regression (Gaussian Assumption):-

In regression problems, we assume that the output variable y follows a Gaussian (Normal) distribution given an input x . Mathematically, this assumption is written as:

$$p(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-h(x))^2}{2\sigma^2}}$$

where:

$h(x)$ is the model function (e.g., linear regression

function $h(x) = \beta_0 + \beta_1 x$)

σ^2 is the variance of the Gaussian noise.

Log-likelihood in Regression:-

Taking the logarithm of the Gaussian likelihood function:

$$\log p(y|x) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y-h(x))^2}{2\sigma^2}$$

Expectation - Maximization (EM) Algorithm:-

* Used for MLE when data has latent variables (hidden factors).

Ex: Gaussian Mixture Model (GMM), where data is generated by multiple Gaussians.

EM steps:-

- 1) E-step:- Estimate expected PDF and parameters for each latent variable.
- 2) M-step:- Optimize parameters using the MLE function.

This process repeats iteratively to refine parameters.

Non-Parametric Density Estimation:-

* Does not assume a specific distribution.

Parzen Window Estimation:-

Given n samples, let k be the no of points in a region R :

$$p = \frac{k}{n}$$

Density estimate:

$$p(x) = \frac{k}{nV}$$

where V is the volume of the region. If R is a hypercube with length h , then:

$$V = h^2 \text{ (for 2D)}, \quad V = h^3 \text{ (for 3D)}$$

the Parzen window function:

$$\phi\left(\frac{x-x_i}{h}\right) = \begin{cases} 1, & \text{if } |x-x_i| < h/2 \\ 0, & \text{otherwise} \end{cases}$$

the probability density function estimate:

$$p(x) = \frac{k}{nV} = \frac{1}{n} \sum_{i=1}^n \frac{1}{V} \phi\left(\frac{x-x_i}{h}\right)$$

Using a Gaussian function instead of a sample window gives the Gaussian Density Function.

K-Nearest Neighbours (KNN) Density Estimation:-

- * Uses k nearest neighbours instead of fixed region size.
- * Density estimate is the average of values returned by k -nearest neighbours.

Feature Engineering and Dimensionality Reduction Techniques:-

* Feature Engineering:-

Selecting or transforming features to improve model performance.

→ Feature Transformation:- Creating new features (eg., BMI from height & weight)

→ Feature Selection:- Choosing relevant features while removing unnecessary ones.

* Feature Subset Selection:-

Reduces dataset size by removing irrelevant and redundant features.

→ Avoids the curse of dimensionality, where too many features increase complexity.

→ Uses a greedy search approach to find the best subset.

Feature removal criteria:-

1. Relevancy:- Important features contribute more (eg., mole vs nose in face detection).
2. Redundancy:- Features providing duplicate info can be removed.
Ceg., age vs date of birth)

Selection Process:-

1. Generate all possible subsets
2. Evaluate subsets and model performance
3. Select the best subset.

Selection methods:-

- * Filter-based:- Uses statistical measures (correlation, mutual information)
- * Wrapper-based:- Uses classifiers for feature selection, offering better performance but higher computation.

Algorithms:-

1) Stepwise Forward Selection:-

- * Starts with no attributes and adds the best one at each step until a good set is formed.

2) Stepwise Backward Elimination:-

- * Starts with all attributes and removes the worst one at each step until a good set is formed.

Note:- Combined approach: Uses both methods - adds the best attribute and removes the worst one.

3) Principal Component Analysis (PCA):-
*) PCA reduces dimensions by transforming data into new features with maximum variance.

Steps:-

- i) Subtract mean from dataset
- ii) Compute covariance matrix.
- iii) Calculate eigenvalues & eigenvectors.
- iv) Select top eigenvectors for transformation.
- v) Transform dataset using selected eigenvectors.

4) Linear Discriminant Analysis (LDA):-

*) LDA reduces dimensions while preserving class separability.

Steps:-

- i) Compute class means.
- ii) Compute b/w-class and within-class scatter matrices.
- iii) Find eigenvectors and eigenvalues.
- iv) Transform data using top eigenvectors.

5) Singular Value Decomposition (SVD):-

SVD is a decomposition technique that represents a matrix A as

$$A = USV^T$$

where,

*) U and V are orthogonal matrices

*) S is a diagonal matrix with singular values.

Steps:-

- i) Compute AA^T

- i) Find its eigenvalues and eigenvectors to form matrix U .
- ii) Sort eigenvalues in descending order.
- iii) Take the square root of eigenvalues to form diagonal matrix S .
- iv) Find eigenvectors and eigenvectors of $A^T A$ to get matrix V .

Q) Let the data points be $\begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$. Apply PCA and find the transformed data.

Sol: Let $X_1 = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$, $X_2 = \begin{bmatrix} 1 \\ 7 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 2 & 1 \\ 6 & 7 \end{bmatrix}$

$$\mu = \text{mean} = \begin{bmatrix} (2+1)/2 \\ (6+7)/2 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 6.5 \end{bmatrix}_{1 \times 2} = \begin{bmatrix} 1.5 & 1.5 \\ 6.5 & 6.5 \end{bmatrix}_{2 \times 2}$$

$$X_{\text{center}} = \begin{bmatrix} 2 & 1 \\ 6 & 7 \end{bmatrix} - \begin{bmatrix} 1.5 & 1.5 \\ 6.5 & 6.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

$$\text{Covariance} = \frac{1}{n-1} (X_{\text{center}} \cdot X_{\text{center}}^T)$$

n : no of features

(here $n=2$)

$$= \frac{1}{2-1} \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

$$X_{\text{cov}} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

To find eigen values (λ), solve $\det(X - \lambda I) = 0$.

$$\Rightarrow \left| \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0.$$

$$\begin{vmatrix} 0.5 - \lambda & -0.5 \\ -0.5 & 0.5 - \lambda \end{vmatrix} = 0$$

$$(0.5 - \lambda)^2 - (0.5)^2 = 0$$

$$0.5^2 + \lambda^2 - 2 \cdot \lambda \cdot 0.5 - (0.5)(0.5) = 0$$

$$\lambda^2 - \lambda = 0$$

$$\lambda(\lambda - 1) = 0$$

$\lambda = 0, 1$ are the eigen values.

Use these λ values, to get eigen vectors.

$$\Rightarrow \det(X - \lambda I) \cdot \bar{V} = 0 \quad (\text{for } \lambda = 1)$$

$$\left(\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & -0.5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -0.5V_1 - 0.5V_2 \\ -0.5V_1 - 0.5V_2 \end{bmatrix} = 0 \Rightarrow \begin{aligned} -0.5V_1 - 0.5V_2 &= 0 \rightarrow \textcircled{1} \\ -0.5V_1 - 0.5V_2 &= 0 \rightarrow \textcircled{2} \end{aligned}$$

$$\therefore \textcircled{1} \Rightarrow -0.5V_1 - 0.5V_2 = 0$$

$$0.5V_1 = -0.5V_2$$

$$V_1 = -V_2 \Rightarrow \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad (\text{or}) \quad \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\therefore$ eigen vector for $\lambda=1$ $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

My for $\lambda=0$.

$$\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$0 = \begin{vmatrix} 0.5 & \lambda - 0.5 \\ \lambda - 0.5 & 0.5 \end{vmatrix}$$

$$\begin{bmatrix} 0.5v_1 - 0.5v_2 \\ -0.5v_1 + 0.5v_2 \end{bmatrix} = 0$$

$$0 = (0.5)(0.5) - (\lambda - 0.5)(\lambda - 0.5)$$

$$0 = (0.5)(0.5) - 0.5 \cdot 0.5 \lambda^2 - \lambda + 0.5$$

$$\Rightarrow 0.5v_1 - 0.5v_2 = 0 \rightarrow \textcircled{3}$$

$$0 = \lambda^2 - \lambda$$

$$-0.5v_1 + 0.5v_2 = 0 \rightarrow \textcircled{4}$$

$$0 = (\lambda - 1)\lambda$$

$$\textcircled{3} \Rightarrow 0.5v_1 - 0.5v_2 = 0$$

$$0.5v_1 = 0.5v_2$$

$$\frac{v_1}{v_2} = 1$$

$$(1 = \lambda \text{ for } \lambda=1)$$

$$0 = \bar{v} \cdot (I\lambda - X)$$

$\therefore$ eigen vector for $\lambda=0$, $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Normalization of eigen vectors,

Note!

In general,

$$e_n = \begin{bmatrix} \frac{x_1}{\sqrt{x_1^2 + x_2^2}} \\ \frac{x_2}{\sqrt{x_1^2 + x_2^2}} \end{bmatrix}$$

$$0 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

$$0 = \begin{bmatrix} 0.5v_1 - 0.5v_2 \\ -0.5v_1 + 0.5v_2 \end{bmatrix}$$

Now $e_1 = \begin{bmatrix} \frac{1}{\sqrt{1^2 + (-1)^2}} \\ \frac{-1}{\sqrt{1^2 + (-1)^2}} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$

$$0 = 0.5v_1 - 0.5v_2 \Rightarrow \textcircled{1}$$

$$0.5v_1 - 0.5v_2 = 0$$

$$0.5v_1 = 0.5v_2$$

$$\text{by } e_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

combine e_1 and e_2 and name it A .

$$\therefore A = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Transformed data:-

$$y = A * X_{\text{centered.}}$$

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ -0.707 & 0.707 \end{bmatrix}$$

ignore 0 ~~now~~

$$y = [-0.707 \quad 0.707]_{1 \times 1}$$

Verify it is correct, $X = A^T y + \text{Mean}$.

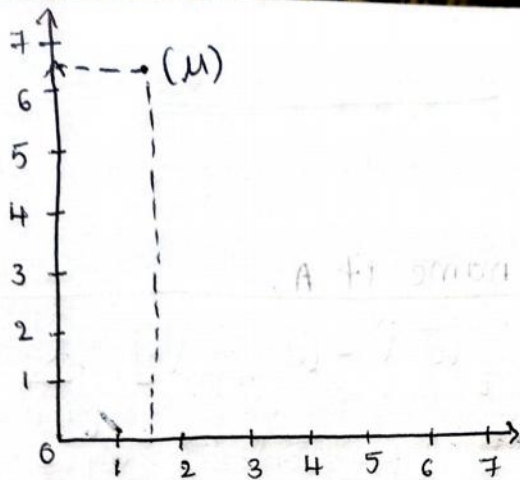
$$X = \left(\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ -0.707 & 0.707 \end{bmatrix} \right) + \begin{bmatrix} 1.5 & 1.5 \\ 6.5 & 6.5 \end{bmatrix}$$

$$= \begin{bmatrix} 1.999 & 1.001 \\ 6.0001 & 6.9998 \end{bmatrix} \approx \begin{bmatrix} 2 & 1 \\ 6 & 7 \end{bmatrix}$$

Verified.

Plotting:-

Step 1:-

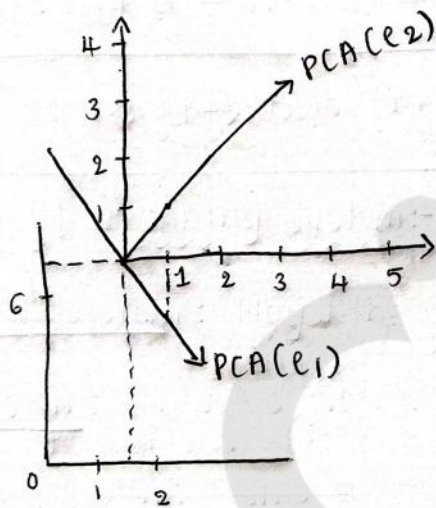


$$\begin{bmatrix} \sigma_1^2 & \sigma_1 \sigma_2 \\ \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}$$

A 2x2 matrix is symmetric

$$\begin{bmatrix} \sigma_1^2 & \sigma_1 \sigma_2 \\ \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix} = A$$

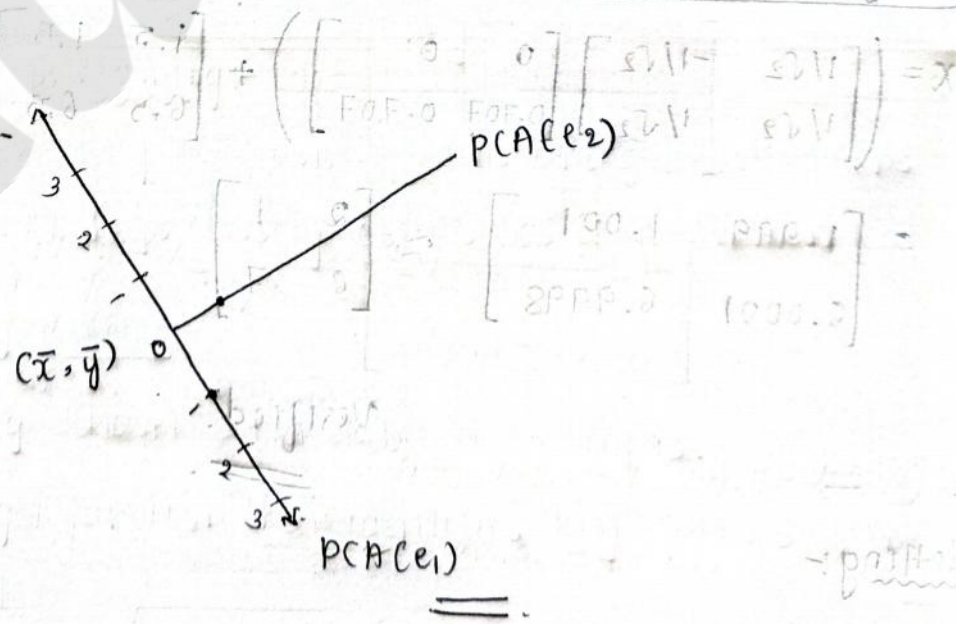
Step 2:-



$$\begin{bmatrix} 1.0 & -0.2 \\ 0.0 & 1.0 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & \sigma_1 \sigma_2 \\ \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix} =$$

Note:-
PCA(e1) and PCA(e2)
are always $\perp$ to
each other.

Final:-



$$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} = P$$

$$A P + P^T A = X$$

$$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} + \begin{bmatrix} 0.0 & 0.0 \\ 0.0 & 0.0 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & \sigma_1 \sigma_2 \\ \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix} = X$$

$$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} =$$

PCA(e1)

Design of a Learning System:-

A learning system is built around a learning algorithm and contains these steps:-

- 1) Choosing a training experience
- 2) Choosing a target function
- 3) Representing a target function
- 4) Function approximation.

→ Training Experience:-

- *> In direct experience, correct moves in a chess game are given.
- *> In indirect experience, only move sequences and results are given.
- *> The system learns better when training and testing samples have the same distribution.

→ Determine the Target Function:-

- *> Defines what the system needs to learn.
- *> The best move is selected based on its effectiveness.
- *> In indirect experience, all legal moves are scored, and the highest scoring move is chosen.

→ Determine the target function representation:-

- *> The knowledge can be stored in tables, rules or neural networks.
- *> A simple linear function can be:

$$V = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

Where, $x_1, x_2, x_3 \rightarrow$ board features

$w_0, w_1, w_2, w_3 \rightarrow$ weights.

choosing an approximation algorithm:-

i) The goal is to adjust weights to minimize error.

* The error is calculated as:

$$E = \sum_{\text{Training samples}} [V_{\text{train}}(b) - \hat{V}(b)]^2$$

* Weights are updated using:

$$w_i = w_i + \mu * \text{error}(b) * x_i$$

where, μ controls the update size.

Components of learning system:-

i) Performance System: Plays the game

ii) Critic System: Generates training samples

iii) Generalizer System: Forms a hypothesis

iv) Experimenter System: Creates a new system based on learned data.

Introduction to Concept Learning:-

Concept learning is a way to learn patterns and generalizations from training data. It helps classify objects based on common features.

* Concept Learning process:-

i) Input:- Training dataset with labeled examples.

ii) Output:- A target function that identifies key features of the concept

iii) Test:- New instances are tested to see if they match the concept.

Ex: Identifying an elephant using features like tusks, fur, size and color.

Representation of a hypothesis:-

A hypothesis (h) is an approximation of a target function that maps input attributes to output values.

* Hypothesis Conditions:-

- "?" means any value is allowed.
- "∅" means no value is allowed.
- Specific values like "Tail = short" define strict conditions.

Example Hypothesis for identifying an elephant:-

Horns	Tail	Tusks	Paws	Fur	Color	Hooves	Size
< No	?	Yes	?	?	Black	No	medium >

this means an elephant must have "no horns", "tusks" and "black color" while other attributes can vary.

Hypothesis space:-

* the hypothesis space includes all possible hypotheses for a target function.

→ version space: The set of hypotheses that are consistent with training data.

→ Example: With 8 attributes, 384 distinct instances can be formed.

→ Challenges: Searching for the best hypothesis efficiently is key.

Heuristic Space Search:-

*> Heuristic search finds an optimized hypothesis using techniques like:

→ Hill climbing

→ Genetic algorithms

→ Best-first search

→ A* algorithm.

*> It improves efficiency by selecting better hypotheses iteratively.

Generalization and Specialization:-

Generalization: This learning methodology will search through the hypothesis space for an approximate hypothesis by generalizing the most specific hypothesis.

Specialization: This learning methodology will search through the hypothesis space for an approximate hypothesis by specializing the most general hypothesis.

Hypothesis space search by Find-S algorithm:-

Algorithm:-

Input: Positive instances in the training dataset

Output: Hypothesis 'h'

1) Initialize 'h' to the most specific hypothesis.

$h = \langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$

2) Generalize the initial hypothesis for the first positive instance

3) For each subsequent instances:

If it is a positive instance:

check for each attribute value in the instance with the hypothesis 'h'.

If the attribute value is the same as the hypothesis value, then do nothing.

Else if the attribute value is different than the hypothesis value, change it to '?' in 'h'.

Else if it is a negative instance,

Ignore it.

Q) Write Find-s algorithm & apply the same for the given dataset.

Location	Price	Size	Conduction	Buy house
Urban	Expensive	Large	New	Yes
Suburban	Moderate	Medium	Old	No
Urban	Moderate	Small	New	Yes
Urban	Moderate	Large	Old	Yes

Sol: Let $h = \langle \phi \quad \phi \quad \phi \quad \phi \rangle$

for $I_1 \Rightarrow h_1 = \langle \text{urban} \quad \text{expensive} \quad \text{large} \quad \text{new} \rangle$

for $I_2 \Rightarrow$ ignore it

for $I_3 \Rightarrow h_3 = \langle \text{urban} \quad ? \quad ? \quad \text{new} \rangle$

for $I_4 \Rightarrow h_4 = \langle \text{urban} \quad ? \quad ? \quad ? \rangle$

$\therefore$ final hypothesis = $\langle \text{urban} \quad ? \quad ? \quad ? \rangle$

Candidate Elimination:-

Algorithm:-

Input: Set of instances in the Training dataset

Output: Hypothesis G and S

1) Initialize G , to the maximally general hypotheses.

2) Initialize S , to the maximally specific hypotheses.

→ Generalize the initial hypothesis for the first positive instance.

3) For each subsequent new training instance,

→ If the instance is +ve,

* Generalize S to include the positive instance,

Check the attribute value of the +ve instance and S ,

• If the attribute value of +ve instance and S are different,

fill that field value with '?'

• If the attribute value of +ve instance and S are same, then do not change.

* Prune G to exclude all inconsistent hypothesis in G with the +ve instance.

→ If the instance is negative,

* Specialize G to exclude the negative instance.

Add to G all minimal specializations to exclude the -ve example and be consistent with S .

• If the attribute value of S and the -ve instance are different, then fill that attribute value with S value.

• If the attribute value of S and negative instance are same, no need to update ' G ' and fill that attribute with '?'

*> Remove from S all inconsistent hypothesis with the negative instance.

Q) Given,

Example	Shape	Size	Color	Surface	Thickness	Target concept
1	Circular	large	light	smooth	thick	+ve.
2	circular	large	light	irregular	thick	+ve
3.	oval	large	dark	smooth	thin	-ve
4.	oval	large	light	irregular	thick	+ve

Solt $S = \langle \emptyset \quad \emptyset \quad \emptyset \quad \emptyset \quad \emptyset \rangle$

$S_1 = \langle \text{circular} \quad \text{large} \quad \text{light} \quad \text{smooth} \quad \text{thick} \rangle$

$S_2 = \langle \text{circular} \quad \text{large} \quad \text{light} \quad ? \quad \text{thick} \rangle$

$S_3 = \langle \text{circular} \quad \text{large} \quad \text{light} \quad ? \quad \text{thick} \rangle$

$S_4 = \langle ? \quad \text{large} \quad \text{light} \quad ? \quad \text{thick} \rangle$

$G_4 = \langle ? \quad ? \quad \text{light} \quad ? \quad ? \rangle, \langle ? \quad ? \quad ? \quad ? \quad \text{thick} \rangle$

$G_3 = \langle ? \quad ? \quad \text{light} \quad ? \quad ? \rangle, \langle ? \quad ? \quad ? \quad ? \quad \text{thick} \rangle$

$\langle \text{Circular} \quad ? \quad ? \quad ? \quad ? \rangle$

$G_2 = \langle \text{circular} \quad ? \quad ? \quad ? \quad ? \rangle, \langle ? \quad ? \quad \text{light} \quad ? \quad ? \rangle, \langle ? \quad ? \quad ? \quad \text{irregula}$

$? \rangle, \langle ? \quad ? \quad ? \quad ? \quad \text{thick} \rangle$

$G_1 = \langle ? \quad ? \quad ? \quad ? \quad ? \rangle$

$G = \langle ? \quad ? \quad ? \quad ? \quad ? \rangle$

$G = \langle ? \quad ? \quad ? \quad ? \quad ? \rangle$

Version space:- (VS)

$S_4 = \langle ? \text{ large light ? thick} \rangle$

$G_4 = \langle ? ? \text{ light ? ?} \rangle, \langle ? ? ? ? \text{ thick} \rangle$

$VS = \langle ? \text{ large light ? ?} \rangle, \langle ? \text{ large ? ? thick} \rangle$

$\langle ? ? \text{ light ? thick} \rangle$

total $3 + 3 = 6$

Modeling in ML:-

A ML model is an abstraction of data that predicts new outcomes based on training. Training involves feeding data to the algorithm to learn patterns, tuning it for performance, validating it, and making predictions on unseen data.

Key concepts:-

1) Parameters in ML:-

→ Model parameters:- learned from data (eg., weights in neural networks)

→ Hyperparameters:- Set before training (eg., no of decision trees in random forests).

2) Model Training & Evaluation:-

→ Dataset split:- Training data trains the model, test data evaluates performance.

→ Training error:- Error on training data.

→ Test error: Error on unseen data.

→ Loss function: Measures prediction accuracy (eg, MSE)

↓
Mean squared error.

ML process:-

- 1) Choose an algorithm
- 2) Train the model on data
- 3) Tune parameters for better accuracy
- 4) Evaluate performance.

Model Selection and Evaluation:-

* Performance: How well the model fits training data.

* Complexity: More complex models may overfit.

Re-sampling Methods (Improving Model Selection):-

* Cross-validation: Splitting data into subsets to train and validate models.

* Hold out method: One-time split into training and test sets.

* K-fold cross validation: Data is split into k parts, training is done $(k-1)$ times, and testing on the remaining data.

* Stratified K-fold Cross validation: Ensures each fold maintains the same class proportion.

* Leave-One-Out Cross-Validation (LOOCV): Each data point is tested once, using the rest for training.

Model Performance:-

Contingency Table for classifier evaluation:-

A classifier model is evaluated using a contingency table, which helps assess its accuracy.

<u>Test vs disease</u>	<u>Has Disease</u>	<u>No Disease</u>
Positive Test	True positive (TP)	False positive (FP)
Negative Test	False negative (FN)	True negative (TN)

Terms:-

- *> TP:- correctly detects disease in sick patients
- *> TN:- correctly detects no disease in healthy individuals
- *> FP:- Wrongly detects disease in a healthy person.
- *> FN:- Fails to detect disease in a sick person.

Metrics Derived from Contingency Table:-

- i) Sensitivity:- The sensitivity of a test is the probability that it will produce a true positive result when used on a test dataset.

$$\text{sensitivity} = \frac{TP}{TP + FN}$$

- ii) Specificity:- The specificity of a test is the probability that a test will produce a true negative result when used on test dataset.

$$\text{Specificity} = \frac{TN}{TN+FP}$$

3) Positive predictive value:- The +ve predictive value of a test is the probability that an object is classified correctly when a +ve test result is observed.

$$PPV = \frac{TP}{TP+FP}$$

4) Negative Predictive value: The -ve predicted value of a test is the probability that an object is not classified properly when a -ve test result is observed.

$$NPV = \frac{TN}{TN+FN}$$

5) Accuracy: The accuracy of the classifier can be shown in terms of sensitivity computed as:

$$\frac{TP+TN}{TP+TN+FP+FN}$$

6) Precision:- Precision is defined as the ratio of TP divided by sum of TP and FP.

$$\text{Precision} = \frac{TP}{TP+FP}$$

7) Recall Same as sensitivity.

$$\text{Recall} = \text{sensitivity} = \frac{TP}{TP + FN}$$

$$PPV = \frac{TP}{TP + FP}$$